

# Ideas Parked — Not Yet Merged (v12)

Staging document; each entry marks its eventual destination

v11 compared the paper against this staging document and found gaps, stale symbols, and one naming collision. This version is a different kind of check, requested separately: a full internal audit of `PA_ReWrite_V15.22.tex` against itself — every cross-reference individually verified, every use of the new verb names checked for drift, and the document read for whether it still reads as one cohesive statement now that Collapse carries full, axiom-level standing. Everything found was fixable directly in the paper, not a conceptual gap needing to be parked; the fixes are recorded here for the historical trail, and the result is `PA_ReWrite_V15.23.tex`.

## 1 Link audit: clean, one structural note

Every label in the document (96) and every reference usage (198 individual call sites, not just unique labels) was checked — not only that each resolves, which the compiler already guarantees, but that the target’s own content actually matches the claim the surrounding sentence makes about it. Zero mismatches found. Zero orphaned references, zero duplicate labels.

One structural note, not a defect: `eq:lexicon_cast` is used in Section 3.1 (Bundle, path, result) before it is defined in Section 6 (Several lexicons, one concept sphere) — a genuine forward reference, correctly resolving to the right equation, just pointing ahead of where it is stated. Checked against the original source: this forward reference predates the terminology rewrite and the Section 2/3 restructuring entirely: this exact sentence, in this exact relative position, forward-references the same concept in the un-rewritten paper. Not introduced by anything done here; not a broken link; recorded only so it is not mistaken for a side effect of the restructuring later.

## 2 Terminology sweep: two stragglers found and fixed

Beyond what earlier passes already caught (the systematic Collapse noun/verb fix, the Five-to-Six-verbs correction, the Table 2 column):

- **One more “a Collapse” noun-phrase**, missed in the earlier sweep, sitting in the Abstract itself: “anti-phase cancellation to black is that reading, not a Bundle or a Collapse.” Same ambiguity as the six instances fixed before — a reader could read “a Collapse” as Axiom 7’s structure rather than the verb. Corrected to “not what Bundle or Collapse would do,” preserving the parallel contrast without the noun-phrase risk.
- **One retired verb still active**, in the Forest costume: “ $\mathfrak{U}$  gathers them into one bowl.” “Gather” was retired as Bundle’s name specifically to stop being used as the verb describing this operator’s own action; this instance described exactly that action with exactly that retired verb. Corrected to “ $\mathfrak{U}$  Bundles them.”

Confirmed zero remaining instances, checked directly: “Gather”/ “gather” (any case), “MixMerge,” the old  $\oplus/\ominus$  assignment, any bare noun-phrase use of “Collapse.” The verb inventory is now used consistently in every one of its roughly 50 occurrences across the document, not merely in the sections originally rewritten.

### 3 Cohesion: the document’s own summaries of itself, checked against what it now contains

This is the one real category of finding — not a conceptual gap, but the document’s own self-description trailing behind its own content. Collapse has a full section, a proven non-idempotence result, a column in the operator-legality table, and a place in the verb-count headline. Three separate places where the paper summarizes “what this does,” written or last touched before Collapse reached that standing, still described only two ways of combining units where the current paper has three:

- The Abstract’s own pipeline sentence — “Bundle  $\rightarrow T \rightarrow$  result is one pipeline; assemble  $\rightarrow$  rest is the other” — silently omitted Collapse. Added: “Collapse its count-preserving alternative when every contribution must survive intact.”
- The Conclusion’s closing summary — “What was written is two pipelines on the same unit” — described Bundle $\rightarrow T$  and Assembly only. Added the same Collapse clause at the equivalent point.
- The Scope-of-the-claim paragraph, a third occurrence of materially the same two-way summary (“through a path... or by letting external seats rest under assembly”), had the identical gap. Fixed the same way, for consistency with the two above.

One further cohesion gap, different in kind: the section that actually *proves* UnMix’s impossibility (“Rotational transposition, and what division is not”) never once says the word “UnMix” in its own text — it is reachable from UnMix’s own section by a forward reference, but carries nothing pointing back. A reader arriving at the proof directly, rather than by reading linearly from the verb table, would have no way to know this is the same fact. Added the name and symbol directly into the proof section’s opening sentence.

Last, Future Work’s own roundup of open items — three named paragraphs, each starting “Open:” — did not include Collapse’s own already-stated open question (Collapse across differing sorts, fully stated where Collapse itself is defined) even though the other three open items from elsewhere in the paper are all gathered there. Added a short fourth paragraph, pointing back to the full statement rather than repeating it, so the roundup does not silently omit it.

### 4 What this review did not find

No new conceptual gap, beyond the three already filed in Section 4 of v11 (Collapse across differing sorts, Collapse as  $T$ ’s actor, Assembly’s silence on Collapse). This pass was looking for a different failure mode — the document disagreeing with itself, not the document being silent on a question — and on that specific question, the answer is that it does not, once the five fixes above are applied. No internal contradiction was found between any two stated properties of Bundle, Remove, Collapse, UnCollapse, Mix, or UnMix; no axiom’s content was found to assume a Collapse behavior inconsistent with the one actually proven; no section written before the terminology rewrite was found using a retired term to make an active claim, beyond the two instances in Section 2 above.